

36 INCHES

THE SIZE OF THE LICK TELESCOPE LENS, THE LARGEST REFRACTOR AT THE TIME

Applications of alternating current electricity were being developed by Serbian-American inventor **Nikola Tesla** (1856–1943). In 1888, he patented the **induction motor**. This is a two-phase machine that uses two alternating currents to produce the rotating magnetic field that makes the rotor turn. The patent was bought

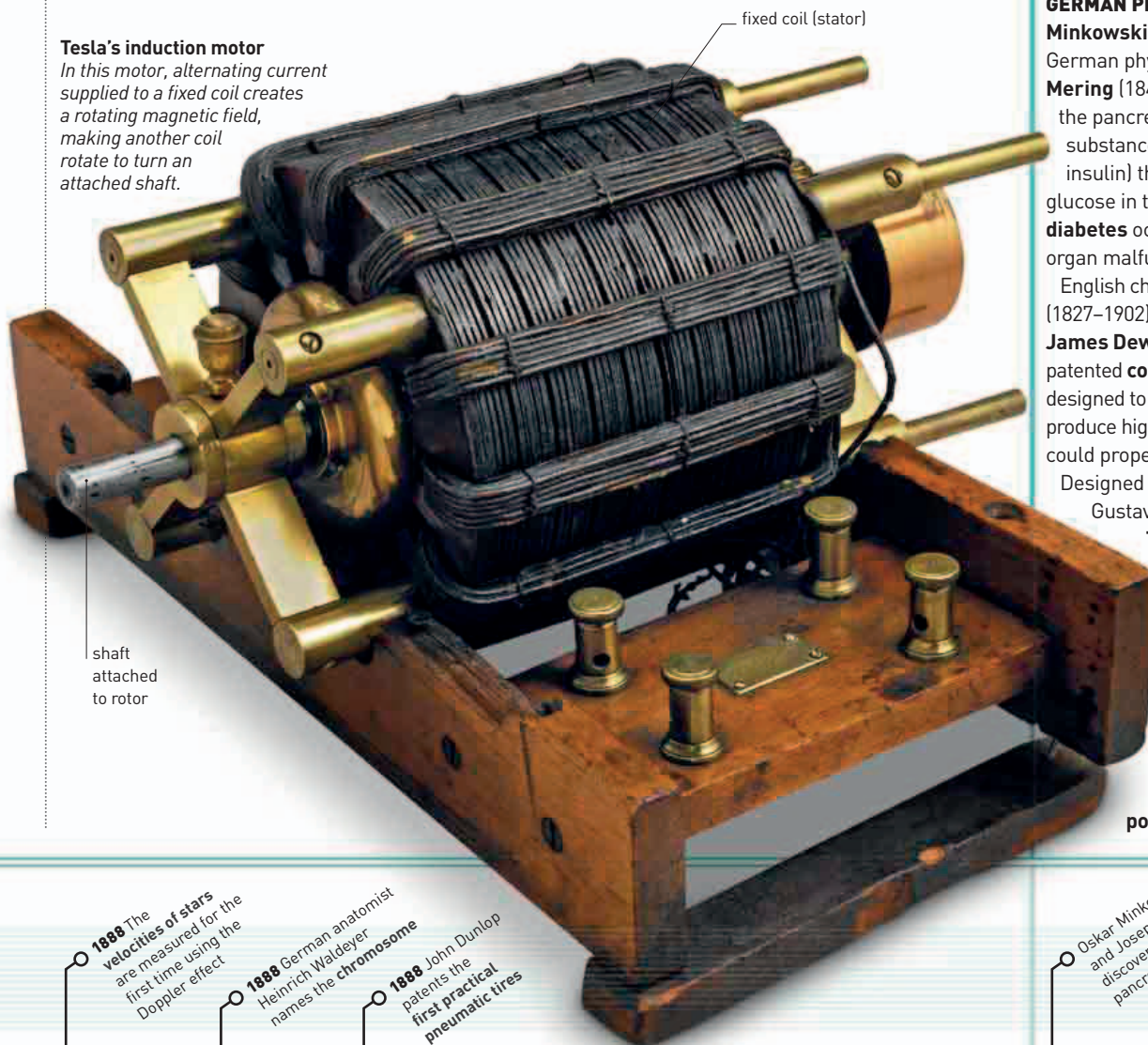
by the Westinghouse Electric Company and used to develop motors, which were widely used in industry and household appliances around the world.

Scottish inventor **John Boyd Dunlop** (1840–1921) developed a **pneumatic bicycle tire** (a tire filled with air) in 1887 and patented it in England in 1888.

The patent was later declared invalid because the principle had been patented earlier by another Scotsman, Robert Thomson (1822–73), in France in 1846 and the US in 1847. However, Dunlop's were the **first practical pneumatic tires**.

Tesla's induction motor

In this motor, alternating current supplied to a fixed coil creates a rotating magnetic field, making another coil rotate to turn an attached shaft.



The Eiffel Tower, Paris, was the tallest man-made structure in the world until the Chrysler Building was erected in New York in 1930.

14 MILES

THE LENGTH OF THE FIRST LONG-DISTANCE POWER TRANSMISSION LINE IN THE US

GERMAN PHYSIOLOGIST Oskar Minkowski (1858–1931) and German physician **Joseph von Mering** (1849–1908) showed that

the pancreas produces a substance (later identified as insulin) that regulates sugar glucose in the body, and that **diabetes** occurs when this organ malfunctions.

English chemist **Frederick Abel** (1827–1902) and Scottish chemist **James Dewar** (1842–1923) patented **cordite**—an explosive designed to burn vigorously and produce high-pressure gases that could propel bullets and shells.

Designed by French engineer Gustave Eiffel, the **Eiffel**

Tower in Paris, France, was opened on March 31. At 984 ft (300 m), the tower was the tallest building in the world at the time.

On June 3, the **first long-distance power transmission**

line in the US was completed. It was installed in Oregon, between a string of lights in Portland and a generator at Willamette Falls.

Russian physiologist **Ivan Petrovich Pavlov** (1849–1936) began studying **conditioning in dogs** in 1889. He had noticed that dogs would begin salivating when they saw the lab technician who fed them. Pavlov began to signal their feeding with the sound of a metronome; soon, the dogs began to salivate each time they heard the metronome (see 1907).

Irish physicist **George FitzGerald** (1851–1901) published a paper suggesting that if all moving objects shrank in the direction of their motion, the results of the Michelson–Morley experiment could be explained. This speculation was based on the idea that electromagnetic forces would squeeze the moving objects. Dutch physicist **Hendrik Lorentz** (1853–1928) came up with a similar idea as well; this shrinking emerges naturally from special theory of relativity (see 1905).

1888 The velocities of stars are measured for the first time using the Doppler effect

1888 German anatomist Heinrich Waldeyer names the chromosome

1888 John Dunlop patents the first practical pneumatic tires

1888 American inventor William Seward Burroughs is granted patents for his adding machine

1888 American entrepreneur and inventor George Eastman markets the first Kodak cameras

October 14, 1888 Louis Le Prince shoots what is possibly the first movie ever made

October 30, 1888 American inventor John Loud is granted a patent for the ballpoint pen

1888 English polymath Francis Galton discovers and names the statistical property correlation

March 12 American inventor Almon Strowger files a patent for an automatic telephone exchange

March 31 The Eiffel Tower is opened to the public

June 3 US's first long-distance power transmission line is completed

George FitzGerald publishes *The Ether and the Earth's Atmosphere*

British naturalist Alfred Russel Wallace publishes his book on natural selection—*Darwinism*

Frederick Abel and James Dewar patent cordite

Ivan Pavlov begins his studies of conditioning in dogs